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PERFORMANCE SPECIFICATION

INTERIM TERRAIN DATA (ITD)/PLANNING INTERIM TERRAIN DATA (PITD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.

a. This specification establishes the second edition military specification requirements for the Defense Mapping Agency's (DMA) Interim Terrain Data (ITD) and Planning Interim Terrain Data (PITD). This document specifies the format, content, and product design of ITD and PITD, which are unsymbolized digital data sets.

b. The DMA Terrain Analysis Program is a dynamic program. This document identifies specifications encountered in the production of the ITD and PITD thematic files. Supplementary instructions may need to be generated as this product evolves. Modifications will be handled through Configuration Management procedures.

1.2 Purpose. Conformance to these specifications will assure uniformity of treatment among all mapping and charting elements engaged in a coordinated production and maintenance program for this product.

1.3 Classification. The ITD and PITD data sets are, respectively, based on the level of detail represented in the 1:50,000/1:100,000 scale Tactical Terrain Analysis Data Base (TTADB) and the 1:250,000 scale Planning Terrain Analysis Data

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Director, Defense Mapping Agency, ATTN: ATC, 8613 Lee Highway, Fairfax, VA. 22031-2137 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Base (PTADB). Both ITD and PITD have an enhanced transportation network, and are provided in a standardized digital format. ITD and PITD are portrayals of analyzed attributes of terrain features (both natural and man-made) that are of significance to tactical (ITD) and planning (PITD) military operations.

1.4 Applicability.

a. For the remainder of this document, the term ITD will be used generically to describe both ITD and PITD. Where it is important to distinguish between the two, this document will do so. Likewise, the term TADB will be used generically to describe both TTADB and PTADB.

b. These specifications apply to all ITD produced by the Defense Mapping Agency and those produced for the Defense Mapping Agency as a result of either government contract or unit tasking.

c. These specifications apply to all activities involved in the preparation and maintenance of ITD.

1.5 ITD design.

a. ITD is a product developed to satisfy the armed services short-term and mid-term requirements for digital terrain analysis data.

b. In the case where TADBs are used as the primary source, ITD will reflect the specification current at the time of TADB collection. In all other cases, the currently configured baselined TADB specification will be used.

c. ITD is designed to use the Defense Mapping Agency Feature File (DMAFF) coding scheme (see 2.1.2.a.), and the Standard Linear Format (SLF) for Digital Cartographic Feature Data (see 2.1.1), for data format and structure.

d. ITD is independent of the method of its production. The production methods result in a standard product that meets the requirements of this specification.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

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SPECIFICATIONS - MILITARY

MIL-J-89100 - Joint Operation Graphics Series 1501A (AIR) and 1501 (GROUND) (JOG A/G)

MIL-T-89301 - 1:50,000 Scale Topographic Maps of Foreign Areas

MIL-T-89304 - Tactical Terrain Analysis Data Base (TTADB) Scale 1:50,000/1:100,000

MIL-P-89305A - Planning Terrain Analysis Data Base (PTADB) Scale 1:250,000

STANDARDS - MILITARY

MIL-STD-2413 - Standard Linear Format (SLF) for Digital Cartographic Feature Data.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

a. DMA Feature File (DMAFF).

(Copies of the above are available from the Defense Mapping Agency, ATTN: ATIS, 8613 Lee Highway, Fairfax, VA. 22031-2137.)

b. DMA Technical Manual (DMA TM) 8358.1, Datums, Ellipsoids, Grids, and Grid Reference Systems, DMA Stock No. DMATM83581TEXT.

(Copies of the above are available from the Defense Mapping Agency, Consumer Interface (OCI), 6001 MacAuthur Boulevard, Bethesda, MD 20816-5001.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards) the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subject to first article inspection in accordance with 4.2.

3.2 Accuracy.

3.2.1 Horizontal accuracy. A formal horizontal accuracy for ITD has not been defined by the users of this product.

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3.2.2 Thematic file relationships.

a. ITD thematic files, shall be prepared such that when the files of a given geographic area are registered together (combined/stacked), they shall bear the same geographic relationship to each other that exists in the source from which they were digitized.

b. Common Open Water (COW) bodies are areal drainage features that meet the minimum size requirements for inclusion in the TADB thematic overlays.

(1) COW bodies are common to four thematic files of a given data set (Surface Configuration, Vegetation, Surface Materials, and Surface Drainage).

(2) COW bodies will be digitized once and replicated into the remaining three files. When digitization is from TADB source, the Surface Drainage COW will be the one digitized.

(3) Subsequent processing of the files may result in slight differences in the final shape of the COW bodies on the four files.

3.3 Datum.

3.3.1 Horizontal datum. Horizontal datum of ITD files shall be the current World Geodetic System - 1984 (WGS 84), or a local datum from DMA TM 8358.1 when no conversion to WGS 84 exists and the source material is an existing TADB on the local datum.

3.3.2 Vertical datum. Vertical datum shall be Mean Sea Level.

3.4 Security.

3.4.1 Security classification. The security classification of the products generated by the use of these specifications will be the lowest category practicable. When it is necessary to assign a security classification to the product, it will be accomplished in accordance with established national security procedures.

3.5 Data density levels.

a. ITD/PITD data is collected at a density of detail that approximates that of the TTADB/PTADB (MIL-T-89304/MIL-P-89305A) overlays, respectively. Therefore, normal data collection density is 1:50,000 for the ITD and 1:250,000 for the PITD.

b. Based on its data collection density, if ITD or PITD are to be output in hardcopy form, the appropriate scale for this output is 1:50,000/1:100,000 for ITD and 1:250,000 for PITD. The 1:100,000 ITD output is the result of a 2x scale reduction of a block of four 1:50,000 ITD data collection cells.

3.6 Data set size. The geographic area of the ITD or PITD data set is based on the 1:50,000/1:100,000 or 1:250,000 topographic map sheet lines (MIL-T-89301/MIL-J-89100), respectively.

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3.7 Continuity (adjoining data set match).

a. Each ITD file area joins the adjacent ITD file area to form a continuous data base with no gaps between files. No file area overlap exists between adjacent files.

b. Features crossing file boundaries shall be continuous, i.e., a feature's geographic position which is located on a file boundary is common to all adjacent files. The only exceptions to this rule are when more current source is used and the feature on the ground has changed (e.g. new road), or when the mismatch is due to different TADB specifications. In these cases, there may be a discontinuity along a file boundary.

3.8 Dimensions.

3.8.1 Unit of measure. The Unit of Measure for the ITD/PITD is Metric.

3.8.2 Minimum sizes. The minimum and maximum sizes of features digitized in most of the thematic files are stated in the TADB specifications current at the time of collection. The features may be digitized as points, lines, or areas depending on the measured values from the source.

3.9 Feature and attribute coding system. ITD feature and attribute coding shall be in accordance with the DMAFF reference (see 2.2.2.a.).

3.10 ITD file. ITD will be produced in the DPS SLF format, which provides a standard format for digital cartographic feature data. Refer to the Military Standard for SLF (see 2.2.1) for more detail on SLF format and structure. Appendix O, Implementing Interim Terrain Data (ITD) in 2-D SLF, provides specific guidance for the implementation of ITD.

3.10.1 Magnetic tape media.

a. Physical characteristics - ITD will be distributed on 9 track, 6250 BPI, 1/2 inch magnetic tapes.

b. Magnetic tape label - The magnetic tape label shall be affixed to the side of the magnetic tape. At a minimum the label shall contain:

(1) Name of the type of data (e.g., ITD).

(2) Date and edition of data.

(3) Area identifier.

(4) Production center tape number.

(5) Tape density.

(6) Blocking of data.

(7) Number of records.

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(8) Copyright note. DMA products may be copyrighted in the name of the U.S. Government in foreign countries that are signatories to the Universal Copyright Convention. To claim this protection, a notice shall be placed on the magnetic tape as follows:

©COPYRIGHT (YEAR) BY THE UNITED STATES GOVERNMENT	- 8 pt. Caps
NO COPYRIGHT CLAIMED UNDER TITLE 17 U.S.C.	- 6 pt. Caps

(9) Security classification of the tape contents.

(10) DMA customer help desk note. The following note (preferrably in 8 pt. C/L, but may be in 6 pt., if necessary) shall be shown:

For questions concerning this or other DMA Products or Services, please telephone the DMA Customer Help Desk, at 1-800-455-0899, Commerical 314-260-1236, or DSN 490-1236

c. Refer to the Military Standard for SLF (see 2.2.1) for further information.

3.11 Thematic file sequence.

a. The respective digital ITD files will be referred to as "thematic files".

b. The ITD shall be produced as a set of six segregated thematic files, duplicating the content of the six TADB thematic overlays, with the addition of enhanced transportation.

c. The six segregated thematic files are listed below and will be stored or written to tape in this order:

SURFACE CONFIGURATION (SLOPE)
VEGETATION
SURFACE MATERIALS
SURFACE DRAINAGE
TRANSPORTATION
OBSTACLES

3.12 ITD/PITD features and attributes.

a. Except as noted in paragraphs 3.12 to 3.17, the features and attributes carried in the ITD thematic files, as per Appendix A, are the same as those required by the TTADB and PTADB specifications (MIL-T-89304 and MIL-P-89305A, respectively). See those specifications for feature and attribute definitions, minimum sizes, usage limitations, placement rules, etc.

b. See Appendixes A and B for a listing of the features, feature codes, and their associated attributes, attribute codes, and attribute value meanings allowable for the ITD thematic files.

c. All features in the ITD thematic files will carry an Overlay Category (OVC) attribute code value corresponding to the particular thematic on which it appears. If a feature appears on more than one overlay, i.e., common open water,

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it will have that thematic's particular OVC code in each file in which it appears. OVC attribute values are shown in Appendix B.

3.13 Surface Configuration (Slope). This section provides the basic guidance for the production of the Surface Configuration (Slope) thematic file for ITD.

3.13.1 General slope information.

a. Information contained in this file represents the maximum slope of the surface at each point on the ground, expressed as percent slope (tangent of the slope angle x 100), rather than in degrees. Slope is defined as (1) ground whose surface forms an angle with the plane of the horizon (a natural or artificial incline), or (2) the degree or extent of deviation from the horizontal. Although there are an infinite number of slope values at a given point, the maximum slope is the critical limiting value for tactical military operations.

b. See Appendix A for a listing of features and their attributes permitted.

c. Areal extent. Whereas surface configuration is represented by an areal file, all areas within the data set boundary must be labeled with a feature code. There will be no void areas in the file.

d. All features in the Surface Configuration thematic file will carry the OVC attribute code of "1".

3.13.2 Miscellaneous Surface Configuration features. Unique and significant slope-related features that have not been otherwise described but are deemed to be militarily significant will be collected as DMAFF Miscellaneous Graphic Features (9D010) and described in the ITD SLF text record of the file.

3.14 Vegetation. This section provides the basic guidance for the production of the Vegetation thematic file for ITD.

3.14.1 General Vegetation information.

a. Vegetation features shown include those which:

(1) Provide orientation.

(2) Afford concealment for troops, vehicles or unattended ground sensors.

(3) Present obstacles to cross-country movement.

(4) Serve as landmarks.

(5) Provide other significant land use information with military significance.

b. Whereas the manually produced hard copy TADB Vegetation thematic overlays could be compiled as either one or two overlays, the ITD Vegetation thematic file will be assembled as a single thematic file.

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c. See Appendix A for a listing of features and their attributes permitted for this thematic file.

d. Areal extent. Whereas vegetation is represented by an areal file, all areas within the data set boundary must be labeled with a feature code. There will be no void areas in the file.

e. All features in the Vegetation thematic file will carry the OVC attribute code "2".

f. The identification of features reflect similarities in military significance and not taxonomy.

g. For areas that contain a variety of vegetation categories which are below minimum size specifications, the recommended procedure is to group the area into the most restrictive category (i.e., most conservative in terms of movement and concealment).

h. Whereas previously produced manual hardcopy TADB Vegetation thematic overlays may contain information concerning Vegetation Roughness Factor (VRF) numerical values, this data will not be incorporated into the ITD Vegetation thematic file.

3.14.2 Miscellaneous Vegetation features. Additional unique and significant vegetation features may be encountered which are of importance to military operations. In some geographic settings, features such as isolated trees, small clumps of trees, golf courses, cemeteries, etc., may be of significance. If a unique and significant vegetation feature (not present in the main body of the specification) is encountered on the source, it will be collected as a DMAFF Miscellaneous Graphic Feature (9D010) and described in the ITD SLF text record of the file.

3.15 Surface Materials. This section provides the basic guidance for the production of the Surface Materials thematic file for ITD.

3.15.1 General Surface Materials information.

a. The treatment of surface materials is limited to those parameters of soils and other surface materials identified as significant for tactical military operations.

b. Soil is defined as the unconsolidated material that overlies bedrock.

c. The Unified Soil Classification System (USCS) is the system used to classify all unconsolidated material (soil). This system classifies soils into 15 categories based primarily on grain size (texture), plasticity, and organic matter content. These features are coded to reflect observed occurrences of the above USCS soil types and other attributes including soil depth, moisture content, and surface roughness characteristics.

d. Surface materials consist of soils and a number of other materials including rock outcrops, permanent snowfields, and evaporites found from the surface to a depth of 50cm (approximately 20 inches), with particular emphasis

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on the depth between 15 to 38cm (6 to 15 inches) below the surface. This is generally the critical layer where the rating cone index (an indicator of the soil load bearing capacity) is considered the most significant measure of trafficability.

e. See Appendix A for a listing of features and their attributes permitted for this thematic file.

f. Areal extent. Whereas surface materials are represented by an areal file, all areas within the data set boundary must be labeled with a feature code. There will be no void areas in the file.

g. All features in the Surface Materials thematic file will carry the OVC attribute code of "3".

3.15.2 Miscellaneous Surface Materials features. If a unique and significant surface materials feature (not present in the main body of the specification) is encountered on the source, it will be collected as a DMAFF Miscellaneous Graphic Feature (9D010) and described in the ITD SLF Text record of the file.

3.15.3 Not Evaluated areas.

a. The not-evaluated code may be used in areas of surface materials identified as being disturbed by man (9D020). Examples are towns, cities, railroad yards, airports, etc. Other areas may include extensive slag piles, mine tailings, land fills, garbage dumps, etc., where the USCS coding would be inappropriate.

b. Since the Ground Surface feature (4A010) and the Soil Type Category attribute do not adequately describe these areas, descriptive information for 9D020 features will be stored in the ITD SLF Text record (Miscellaneous Text record).

3.15.4 Surface roughness classification and coding.

a. Surface roughness is synonymous with microrelief and covers the expression of the land surface or surface geomorphic features which are less than the contour interval of the base map in height. Surface roughness is that aspect of the microrelief on the land surface (boulder fields, hummocky ground, gullies, rugged bedrock, etc.) which reduces the rate of cross country movement for vehicles or foot troops.

b. Surface Roughness Qualifier (SRQ=0-98) (attribute/value numbers). Surface roughness is classified and coded in the surface materials thematic file by a project-tailored set of sequential numbers designating the surface roughness type. Each separate surface roughness type found in the project area is assigned a Surface Roughness Qualifier or type number. The only surface material types not assigned surface roughness type numbers are the Not-Evaluated and COW features.

c. Surface Roughness Qualifier (SRQ=0-98) (attribute value meaning descriptors).

(1) Each surface roughness type identified and number coded in the project is given a corresponding surface roughness descriptor.

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(2) The surface roughness descriptors are a set of generalized statements about the small-scale differences in relief (natural and/or cultural) that are not normally shown or interpretable on a regular topographic map. These descriptors shall detail the surface roughness within specific mapping units rather than combining several different and separately occurring surface roughness descriptions together.

(3) The actual surface roughness descriptions associated with each Surface Roughness Qualifier are stored in the ITD SLF text record (Surface Roughness Table).

(4) The surface roughness type numbers 0, 1, and 2 are standardized. Refer to the T/PTADB specifications and MIL-STD-2413, Standard Linear Format (SLF) for Digital Cartographic Feature Data, Appendix O.

(5) The remaining surface roughness type numbers and descriptors (SRQ=03-98) are the analyst tailored types and are formatted as described in MIL-STD-2413, Standard Linear Format (SLF) for Digital Cartographic Feature Data, Appendix O.

(6) Whereas previously produced manual hardcopy TADB Surface Materials thematic overlays may contain information concerning Surface Roughness Factor (SRF) numerical values accompanying each Surface Roughness Qualifier and Description, these SRF numerical values will not be incorporated into the ITD Surface Materials thematic file.

3.16 Surface Drainage. This section provides the basic guidance for the production of the Surface Drainage thematic file for ITD.

3.16.1 General Surface Drainage information.

a. See Appendix A for a listing of features and their attributes permitted for the Surface Drainage thematic file. See TADB specifications for specific inclusion conditions.

b. Linear and areal extent. Whereas Surface Drainage is represented by a combination of feature types (mostly linear with some point and areal features) most of the area within the data set boundary of the covered area is not assigned a feature and/or attribute codes.

c. All features in the Surface Drainage thematic file will carry the OVC attribute code of "4".

3.16.2 Miscellaneous Surface Drainage features. Additional Surface Drainage features may be encountered which are of major significance to military operations, especially river and channel crossings and/or landings. In some environments, features such as intermittent lakes, washes/wadis, anastomosing streams, elevated aqueducts, tidal flats, weirs, features under construction, etc., may be of operational and landmark significance. Unique and significant Surface Drainage features not found in the specification will be shown as DMAFF Miscellaneous Graphic Features (9D010) and described (along with any new measurements made for the features) in the ITD SLF text record of the file.

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3.17 Transportation. This section provides the basic guidance for the production of the Transportation thematic file for the ITD.

3.17.1 General Transportation Information.

a. The features and attributes in this thematic file represent transportation features over which troops and supplies can be moved during a tactical military operation. The transportation thematic file consists of features required in TADB specifications in addition to the enhanced transportation guidelines as outlined in this section.

b. If associated attributes for a feature are unknown, guidelines presented in DMAFF specifications should be followed.

c. Whereas Transportation is represented by a combination of feature types (mostly linear with some point and areal features) most of the area within the data set boundary of the covered area is not assigned a feature and/or attribute codes.

d. See Appendix A for a listing of features and their attributes permitted for the Transportation thematic file.

e. Every feature in the Transportation thematic file will carry the OVC attribute value of "5".

3.17.2 Railroads. Railroad tracks are classified and attributed as a track type, track gauge, number of tracks, and electrification status.

3.17.3 Roads.

a. Enhanced transportation. The following describes the collection density and attribution of roads.

(1) Road features required in the TADB specifications will be portrayed and fully attributed.

(2) All roads attributed on specialized "Road and Bridge" maps, where available, will be digitized and fully attributed as per those sources.

(3) All roads, cart tracks and larger, derived solely from the base map source will be divided into four categories with the following road characteristics as their standardized attributes:

(a) All Weather, Hard Surface Highway/Roads (1P030, OVC=5, WTC=1, RST=1, WDD=60).

(b) All Weather, Loose Surface Roads (1P030, OVC=5, WTC=1, RST=2, WDD=50).

(c) Fair Weather, Loose Surface Roads (1P030, OVC=5, WTC=2, RST=2, WDD=40).

(d) Cart Track (1P010, OVC=5), without further attribution.

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(4) Enhanced transportation will not affect portrayal of road networks in urban areas. Inside urban areas only a representative pattern of roads will be shown. This pattern will include all major through routes.

b. A road segment is a single section of road between two nodes. Road segments carry the same classification and attribute characteristics throughout their length. A road feature is a road segment or segments of a road that carry the same classification and attribute characteristics throughout lengths.

(1) Individual road segments are formed at road junctions (intersections). Road features are formed at points of attribute changes.

(2) Railroads crossing road features form segments of each feature. New features are not formed. A node is placed at the point of intersection.

(3) Point features associated with roads, such as sharp curves, drop gates, etc., do not affect road features. Individual road segments are formed.

(4) Bridges, tunnels, and other features that roads pass over or through do not affect road features as long as the road has the same classification and attributes on both sides. If they are different at each end of the feature, the higher road classification, consistent with the design and structural characteristics of the feature, is considered to cross over or through the feature and then change on the other side.

(5) A road intersection is an at-grade crossing, meeting, or junction of two or more roads. Roads overpassing or underpassing other roads on bridges (or elevated structures) or through tunnels do not affect road features. Individual road features are not formed, unless the classification and/or attributes are different on each side of the feature.

3.17.4 Bridges .

a. Bridges that are required in TADB specifications are portrayed and fully attributed.

b. All road bridges derived solely from specialized "Road and Bridge" maps are to be digitized and fully attributed per that source.

c. Road bridges derived solely from the base map source are portrayed with a unique bridge number (as per 3.17.4d. below). All remaining attributes shall either default to "Unknown" or have valid values entered, if available from source material. All such bridges will be point features as their lengths are unknown.

d. All road bridges on the Transportation thematic file are given a unique integer bridge number. This information is stored in the Bridge Reference Number (BRN).

e. Bridge spans (1Q045) are sections of the bridge between successive supports (i.e., pillars, piers, or abutments). These features are portrayed as a point or a line centered at either the mid-point of its associated bridge feature or at the point where the underpassing feature(s) (e.g. canal, stream, road, railroad, etc.) is beneath it.

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(1) Bridge spans are portrayed for road bridges only.

(2) The Bridge Reference Number (BRN) serves to tie the bridge span information back to the bridge. The associated component bridge spans of a bridge receive the same BRN value as the bridge itself.

(3) Span length is the bridge centerline distance from the intersection point of the load carrying spanning members or surface with the end plate on the abutment or support at one end to the same on the other end. The length is measured in decimeters. This is the span length which must be replaced if the span is removed.

(4) If the bearing to bearing length of spans is the only span length known, it will be shown with a warning note to this effect attached to the ITD SLF text record.

3.17.5 Tunnels.

a. All tunnels along the railroads and roads that can be identified on the source and those along roads that can be derived from the base topographic map are included in this thematic file.

b. The hidden inner road or railroad passing through the tunnel is given an approximate delineation in this thematic file.

3.17.6 Miscellaneous Transportation features. In some geographic settings, unique transportation features may be encountered which are significant to military operations along the transportation network. Such features as route segment vertical lifts, trails, overhead obstructions, restricted passages, snowsheds, canals, culverts, elevated transportation structures, etc., in certain environments and conditions may play a critical role in on-route operations. Unique and significant transportation features found on the source which are not specifically identified in Appendix A, will be shown as DMAFF Miscellaneous Graphic Features (9D010) and described in the ITD SLF text record file.

3.18 Obstacles. This section provides the basic guidance for the production of the Obstacles thematic file for ITD.

3.18.1 General Obstacles information.

a. The treatment of obstacles is limited to any natural and/or man-made features that divert ground based military cross-country movement.

b. As much as possible, obstacles should be considered as independent of vehicle/troop type, (i.e., medium and large tanks, large wheeled vehicles, small wheeled vehicles, small tracked vehicles, and foot troops).

c. See Appendix A for a listing of features and their attributes permitted for this thematic file.

d. Whereas obstacles are represented by linear features (with some point and areal features), most of the area within the data set boundary is not assigned a feature and/or attribute codes.

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e. All features in the Obstacles thematic file will carry the OVC attribute code "6".

3.18.2 Miscellaneous Obstacle features. Additional obstacle features are those features that hinder or obstruct military ground movement. In some geographic settings, features such as shelterbelts, on the ground aqueducts, elevated structures, kanats, wooded gullies, permanent military obstructions such as anti-tank ditches, impact areas, minefields, etc., may be of significance. If a unique and significant obstacle feature is present on the source, it will be collected as a DMAFF Miscellaneous Graphic Feature (9D010) and described in the ITD SLF text record of the file.

3.19 Names and Labeling.

a. In the ITD thematic files, features are normally identified by feature type and/or attribute code value and not by name(s). In the rare case that a miscellaneous or unique feature should need to be named in the thematic file, the name should be taken from the base map to which the digitized thematic overlay would be registered.

b. Names are not normally included in ITD. If they are entered in Text files, they shall be those names approved by the U.S. Board of Geographic Names. Normally, U.S. maps of similar scale serve as a guide to features to be named. Individual features of a group are not labeled, instead the names of groups of features are recorded (e.g., archipelago, mountain range, etc.).

3.20 Reproduction and Storage. The ITD thematic files will be reproduced and stored as 9 track, 6250 BPI, magnetic tapes.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article inspection is required (see 3.1), one complete set of ITD thematic files over an area shall be examined for defects as specified in 4.3.1, and the construction record reviewed for compliance with 4.3.2.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.3.1 and the review of 4.3.2.

4.3.1 Examination. The ITD files shall be examined for defects and errors as specified by the contract or government. Required corrections shall be made to all files and reproducible materials before being sent to the next production stage. Defects detected during the inspection of the reproduced "catch copy" shall be evaluated by DMA for criticality, and suitable corrective action.

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4.3.2 Review of construction records. Records about the construction of the ITD files shall be maintained. The records shall document sources, decisions regarding reconciliation of conflicting data, etc. ITD file records/ construction histories shall be reviewed concurrently with visual examinations (see 4.3.1). to ensure that proper cartographic and data processing procedures have been followed.

4.4 Government furnished material. The contractor shall not duplicate, copy, or otherwise reproduce the MC&G property for purposes other than those necessary for the performance of the contract.

4.5 Government property surplus. At the completion of performance of the contract, the contractor, as directed by the contracting officer, shall either destroy or return to the government all government-furnished MC&G property not consumed in the performance of the contract.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. ITD is a product developed to satisfy the armed services short and mid-term requirements for digital terrain analysis data.

6.2 Acquisition requirement. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of the individual documents referenced (see 2.2.1 and 2.2.2).

c. When a first article is required (see 3.1 and 4.2).

d. Packaging requirements (see 5.1).

6.3 Supersession. These specifications supersede the Military Specifications for Interim Terrain Data (ITD)/Planning Interim Terrain Data (PITD), MIL-I-89014, 30 November 1990.

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6.4 Definitions.

6.4.1 TTADB. The Tactical Terrain Analysis Data Base (TTADB) is a 1:50,000/ 1:100,000 scale geographic information system type data base consisting of a set of selected single subject thematic terrain information overlays used to satisfy tactical military requirements. Data on the physical, biological and cultural features of the Earth's surface is presented in a hard copy cartographic format.

6.4.2 PTADB. The Planning Terrain Analysis Data Base (PTADB) is a 1:250,000 scale geographic information system type data base consisting of a set of selected single subject thematic terrain information overlays used to satisfy planning military requirements. Data on the physical, biological and cultural features of the Earth's surface is presented in a hard copy cartographic format.

6.5 Subject term (key word) listing. This paragraph contains an alphabetical listing of subject terms (key words) that allow for identification of the document during retrieval searches. Note subject terms do not repeat words from title of this document, "Military Specifications, Interim Terrain Data (ITD)/ Planning Interim Terrain Data (PITD)":

Airfields
Bridges/Bridge Spans
Defense Mapping Agency Feature File (DMAFF)
Enhanced Transportation
Landing Areas
Miscellaneous Features
Obstacles
Open Water
Planning Terrain Analysis Data Base (PTADB)
Railroads
Roads
Runways
Soil Moisture
Standard Linear Format (SLF)
Streams
Surface Configuration (Slope)
Surface Drainage
Surface Materials
Surface Roughness Qualifiers/Descriptions
Transportation
Tactical Terrain Analysis Data Base (TTADB)
Thematic File
Tunnels
Vegetation
Unified Soil Classification System (USCS)

6.6 Changes from previous issue. The major change in this edition of the ITD/PITD Military Specification (Mil-Spec), MIL-I-89014A, from the first edition, MIL-I-89014, 30 November 1990, is the inclusion of all the first edition Amendments 1 through 4. Some of these include the deletion of the requirement to generate Vegetation and Surface Roughness Factor (VRF and SRF, respectively) numerical values reflecting the degree of vehicle speed degradation within areas of those thematic types. Other changes involve corrections to the Mil-Spec format itself, addition of copyright note, and some minor corrections to various features and

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attributes. New changes to the text herein include: (1) A new classification note in section 1.3; (2) movement of old security note from section 1.3 to section 3.4, and a renumbering of section 3 from thereon; (3) a reordering and address updating for the documents in 2.2.1 and 2.2.2, as well as changing the SLF reference to its new Military Standard, MIL-STD-2413, and reference updating throughout this document; (4) addition of the DMA Customer Help Desk telephone number note to the tape label and in section 6.9; (5) correction of WDD default values for roads taken from base map (page 11); (6) addition of the word "format" to definitions of TTADB and PTADB; (7) deletion of word "Soils" from key word listing; and (8) replacement of section 5 on packaging with an updated shorter version. Other new changes include, but are not limited to, the following attributes in Appendix B: (1) Maximum attribute value for both BGL and BGR is increased from 100% to 998%; (2) The attribute value of zero (0 = Unknown) is deleted from the allowable values for BRN; (3) The value three (3 = Elevated on Grade/Levee (Earthwork) is deleted from the allowable values for RSC; (4) For WDA, the attribute values 1 and 5 have been corrected to read, "1 <= 0.8 m", and, "5 <= 1.2 m"; (5) the addition of the wording "Appendix B" at the top of each page; (6) the deletion of the zero value (0 = Unknown) for the following attributes: HYC, LTC, RRA, RRC, RST, TUC, TWC, and WTC. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Classification and special handling of thematic files.

a. The classification of the final ITD files will be determined by the appropriate security section responsible for the final classification. The lowest possible classification of the final product is desired.

b. Even though the final thematic files might be unclassified, a handling caveat could be required. Some NATO and other countries have mapping and other agreements which dictate the handling of materials produced over their country. Security elements should check for caveat requirements at the beginning of each project.

6.8 Enhanced user package. In order to allow end users to make more complete analyses of the area covered by the ITD files, when an ITD file is ordered, the user package shall be enhanced to include:

a. Transmittal summary sheet.

b. ITD on magnetic tape.

c. Digital Terrain Elevation Data (DTED) on magnetic tape (shall always be transmitted with ITD).

6.9 DMA customer help desk. For questions concerning this or other DMA products, services, or specifications, please telephone the DMA Customer Help Desk at: 1-800-455-0899, Commercial (314) 260-1236, or DSN 490-1236

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APPENDIX A

ITD/PITD FEATURE AND ATTRIBUTE ORGANIZATIONAL TABLE

A.1 SCOPE

A.1.1 Scope. This appendix presents information about the features and their associated attributes as carried in the ITD/PITD thematic files. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

A.3 ITD/PITD SET UP OF FEATURES AND ATTRIBUTES

A.3.1 Feature and Attribute Organizational Table.

a. This table has six sections, each of which corresponds to and is representative of its associated ITD thematic file. The six section headings are: Surface Configuration (Slope), Vegetation, Surface Materials, Surface Drainage, Transportation, and Obstacles. The miscellaneous feature code (9D010) has been provided for each section, and is available for use in the event that a feature or features are encountered that are not described in this specification. Text descriptions are used to describe the miscellaneous feature(s) in the ITD SLF text record.

b. The table presents information about the ITD features, attributes, and values as folows:

(1) The first column, labeled "F Code", contains the DMA Feature File (DMAFF) code.

(2) The second column labeled (ITD and PITD) is used to indicate which features are required in the ITD/PITD specifications. If the feature is required in only a ITD, then a letter "T" is placed in the column. If the feature is required in only a PITD, then a letter "P" is placed in the column. No entry in this column means that the feature is applicable to both ITD and PITD files.

(3) The third column is labeled "Feature Name" with a designation in brackets "[DMAFF Feature Name]." The first name is the feature name for this item as defined for ITD. The second name located within brackets [ ] is the name for the same item found in DMAFF with that particular feature code number.

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(4) The fourth column labeled "F Type" designates which feature types are allowable for this feature, that is: point, line, or area.

(5) The fifth column labeled "F At. No." is the field attribute number, which is the feature header field (location) in the digital data where this attribute is stored.

(6) The sixth column labeled "At. Code" contains the attribute code. This is the three character alphanumeric designation of the different attribute codes which the particular feature can have.

(7) In the seventh column labeled "Values" are the allowable values that the attribute code can have.

(8) In the eighth column labeled "Attribute" is the name of the attribute code designated in column six.

Section 1 SURFACE CONFIGURATION (SLOPE)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2A040		Open Water (Same)	Area	F-15	OVC	1	Overlay Category
3A060		Slope	Area	F-0	GSC	0-7	Ground Slope Category
				F-15	OVC	1	Overlay Category
9D010*		Miscellaneous Surface Configuration Features (Miscellaneous Graphic Feature)	Point	F-15	OVC	1	Overlay Category
			Line	F-15	OVC	1	Overlay Category
			Area	F-15	OVC	1	Overlay Category

* In the ITD/PITD SLF Text record enter the ground slope percentage range for all Miscellaneous Surface Configuration features.

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Section 2 VEGETATION

F Code	ITD (T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1L020		Built-Up Area [Same]	Area	F-15	OVC	2	Overlay Category
2A040		Open Water [Same]	Area	F-15	OVC	2	Overlay Category
2H090		Wetlands [Same]	Area	F-15	OVC	2	Overlay Category
4A010		Bare Ground [Ground Surface]	Area	F-9 F-15	MCC OVC	4 2	Material Composition Category Overlay Category
5A010		Dry Crops [Cropland (Cultivated)]	Area	F-7 F-15	VEG OVC	1 2	Vegetation Characteristics Overlay Category
5A010		Wet Crops [Cropland (Cultivated)]	Area	F-7 F-15	VEG OVC	4 2	Vegetation Characteristics Overlay Category
5A010		Terraced Crops [Cropland (Cultivated)]	Area	F-7 F-15	VEG OVC	3 2	Vegetation Characteristics Overlay Category
5A010		Shifting Cultivation [Same]	Area	F-7 F-15	VEG OVC	2 2	Vegetation Characteristics Overlay Category

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Vegetation (Continued)

F Code	ITD (T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
5A010	(P)	Agriculture Area with Scattered Forests [Same]	Area	F-7	VEG	5	Vegetation Characteristics
				F-15	OVC	2	Overlay Category
5A040		Orchard/ Plantation, (Deciduous) [Same]	Area	F-7	VEG	0,13	Vegetation Characteristics
				F-10	HGT	0-150	Height of Feature above ground level (meters)
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
				F-24	SDS	0-900	Stem Diameter Size (cm)
				F-25	TSD	0-500	Tree Spacing Category(decim.)
5A040		Orchard/ Plantation, (Coniferous/ Evergreen) [Same]	Area	F-7	VEG	0,14	Vegetation Characteristics
				F-10	HGT	0-150	Height of Feature above ground level (meters)
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
				F-24	SDS	0-900	Stem Diameter Size (cm)
				F-25	TSD	0-500	Tree Spacing Category(decim.)
5A040		Orchard/ Plantation, (Mixed) [Same]	Area	F-7	VEG	0,15	Vegetation Characteristics
				F-10	HGT	0-150	Height of Feature above ground level (meters)
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
				F-24	SDS	0-900	Stem Diameter Size (cm)
				F-25	TSD	0-500	Tree Spacing Category(decim.)

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Vegetation (Continued)

F Code	ITD (T) PITD (P)	Feature Name [DMAFT Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
5A040		Orchard/ Plantation, (Palm) [Same]	Area	F-7	VEG	0,17	Vegetation Characteristics
				F-10	HGT	0-150	Height of Feature above ground level (meters)
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
				F-24	SDS	0-900	Stem Diameter Size (cm)
				F-25	TSD	0-500	Tree Spacing Category(decim.)
5A050		Vineyard/Hops [Same]	Area	F-15	OVC	2	Overlay Category
5B010		Grassland Pasture,Meadow [Herbaceous Area]	Area	F-7	VEG	8	Vegetation Characteristics
				F-15	OVC	2	Overlay Category
5B010		Grassland with scattered trees [Herbaceous Area]	Area	F-7	VEG	9	Vegetation Characteristics
				F-15	OVC	2	Overlay Category
5B020		Brushland/Scrub (Open to Medium) [Shrub/Brush/Scrub]	Area	F-15	OVC	2	Overlay Category
				F-22	BDC	1	Brushland Density Category
5B020		Brushland/Scrub (Medium to Dense) [Shrub/Brush/Scrub]	Area	F-15	OVC	2	Overlay Category
				F-22	BDC	2	Brushland Density Category
5C010		Bamboo/ Wild Cane [Bamboo]	Area	F-15	OVC	2	Overlay Category

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Vegetation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
5C030		Coniferous/ Evergreen Forest [Trees]	Area	F-7	VEG	14	Vegetation Characteristics
				F-10	HGT	0-150	Height of Feature above ground level (meters)
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
				F-24	SDS	0-900	Stem Diameter Size (cm)
				F-25	TSD	0-500	Tree Spacing Category(decim.)
5C030		Deciduous Forest [Trees]	Area	F-7	VEG	13	Vegetation Characteristics
				F-10	HGT	0-150	Height of Feature above ground level (meters)
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
				F-24	SDS	0-900	Stem Diameter Size (cm)
				F-25	TSD	0-500	Tree Spacing Category(decim.)
5C030		Mixed Forest [Trees]	Area	F-7	VEG	15	Vegetation Characteristics
				F-10	HGT	0-150	Height of Feature above ground level (meters)
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
				F-24	SDS	0-900	Stem Diameter Size (cm)
				F-25	TSD	0-500	Tree Spacing Category(decim.)
5C030		Forest Clearing [Trees]	Area	F-7	VEG	24	Vegetation Characteristics
				F-15	OVC	2	Overlay Category
5D030		Marsh/Bog [Marsh]	Area	F-15	OVC	2	Overlay Category

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Vegetation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
5D040		Swamp, Deciduous [Swamp]	Area	F-7	VEG	0,13	Vegetation Characteristics
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
5D040		Swamp, Coniferous/ Evergreen [Swamp]	Area	F-7	VEG	0,14	Vegetation Characteristics
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
5D040		Swamp, Mixed [Swamp]	Area	F-7	VEG	0,15	Vegetation Characteristics
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
5D040		Swamp, Mangrove [Swamp]	Area	F-7	VEG	0,19	Vegetation Characteristics
				F-15	OVC	2	Overlay Category
				F-22	UGD	0,1,2	Undergrowth Density Category
				F-23	DMT	0-100	Density Measure (% tree cover)
9D010	*	Miscellaneous Vegetation Feature [Miscellaneous Graphic Features]	Area	F-15	OVC	2	Overlay Category

* In the ITD/PITD SLF text record enter the complete feature description and dimensional values for all characteristics of the Miscellaneous Vegetation features.

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Section 3 SURFACE MATERIALS

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2A040		Open Water [Same]	Area	F-15	OVC	3	Overlay Category
2J100		Permanent Snowfields [Snowfields, Ice Fields, Ice Caps]	Area	F-6	SRQ	0-98	Surface Roughness Qualifier
				F-15	OVC	3	Overlay Category
4A010		Gravel, Well Graded [Ground Surface]	Area	F-2	STC	1	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Gravel, Poorly Graded [Ground Surface]	Area	F-2	STC	2	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Gravel, Silty [Ground Surface]	Area	F-2	STC	3	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category

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Surface Materials (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
4A010		Gravel, Clayey [Ground Surface]	Area	F-2	STC	4	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Sand, Well Graded [Ground Surface]	Area	F-2	STC	5	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Sand, Poorly Graded [Ground Surface]	Area	F-2	STC	6	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Sand, Silty [Ground Surface]	Area	F-2	STC	7	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category

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Surface Materials (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
4A010		Sand, Clayey [Ground Surface]	Area	F-2	STC	8	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Silt [Ground Surface]	Area	F-2	STC	9	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Organic Silt [Ground Surface]	Area	F-2	STC	11	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Inorganic Silt [Ground Surface]	Area	F-2	STC	13	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category

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Surface Materials (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
4A010		Clays [Ground Surface]	Area	F-2	STC	10	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Fat Clays [Ground Surface]	Area	F-2	STC	12	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Organic Clays [Ground Surface]	Area	F-2	STC	14	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category
4A010		Peat/Organic Soils [Ground Surface]	Area	F-2	STC	15	Soil Type Category
				F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	77	Material Composition Category
				F-15	OVC	3	Overlay Category

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Surface Materials (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
4A010		Evaporites [Ground Surface]	Area	F-3	SDC	0,1,2	Soil Depth Category
				F-4	SWC	0-3	Soil Wetness Category
				F-6	SRQ	0-98	Surface Roughness Qualifier
				F-9	MCC	24	Material Composition Category
				F-15	OVC	3	Overlay Category
4B160		Rock Outcrop [Rock strata, Rock Formation]	Area	F-6	SRQ	0-98	Surface Roughness Qualifier
				F-15	OVC	3	Overlay Category
9D010	*	Miscellaneous Surface Materials (Soils) Feature [Miscellaneous Graphic Features]	Area	F-15	OVC	3	Overlay Category
9D020	**	Not Evaluated [Void Collection Area]	Area	F-15	OVC	3	Overlay Category

* In the ITD/PITD SLF text record enter the complete feature description and dimensional values for all characteristics of the Miscellaneous surface material features.

** In the ITD/PITD SLF text record describe feature.

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Section 4 SURFACE DRAINAGE

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2A030		Island [Same]	Area	F-15	OVC	4	Overlay Category
2A040		Open Water [Same]	Area	F-15	OVC	4	Overlay Category
2H010		Covered Drainage [Aqueduct]	Line	F-4	LOC	0,1	Location/Origin Category
				F-5	ACC	0,1,2	Accuracy Category
				F-15	OVC	4	Overlay Category
2H020	(T)	Canal/Channelized Stream/Irrigation Canal/Drainage Ditch, Narrow [Canal]	Line	F-5	RRC	4	Railroad/Road Drainage Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4	Water Depth Average
				F-18	MCC	0,5,14,35, 57,66,69,76	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	0-45	Gap Width (Decimeters)
2H020		Canal/Channelized Stream/Irrigation Canal/Drainage Ditch, Medium [Canal]	Line	F-5	RRC	7	Railroad/Road Drainage Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14, 35,57 66,69,76 (T)	Material Composition Category

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Surface Drainage (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2H020		Canal/Channelized Stream/Irrigation Canal/Drainage Ditch, Medium (Continued)		F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	46-180 (T)	Gap Width (Decimeters)
				F-38	GWD	181-1420 (P)	Gap Width (Decimeters)
2H020		Canal/Channelized Stream/Irrigation Canal/Drainage Ditch, Wide [Canal]	Area	F-5	RRC	9	Railroad/Road Drainage Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	181-50,000 (T)	Gap Width (Decimeters)
				F-38	GWD	1421-50,000 (P)	Gap Width (Decimeters)
2H055		Float Bridge/ Raft Site [Float/Raft Site]	Point	F-15	OVC	4	Overlay Category
			Line	F-15	OVC	4	Overlay Category

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Surface Drainage (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2H070		Off Route Ford [Ford]	Point	F-15	OVC	4	Overlay Category
			Line	F-15	OVC	4	Overlay Category
2H140	(T)	Intermit/Ephemeral Stream, Narrow [River/Stream]	Line	F-5	RRC	4	Railroad/Road Drainage Category
				F-6	HYC	6	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	0-45	Gap Width (Decimeters)
2H140		Intermit/Ephemeral Stream, Medium [River/Stream]	Line	F-5	RRC	7	Railroad/Road Drainage Category
				F-6	HYC	6	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank

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Surface Drainage (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2H140		Interit/Ephemeral Stream, Medium (Continued)		F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	46-180 (T)	Gap Width (Decimeters)
				F-38	GWD	181-1420 (P)	Gap Width (Decimeters)
2H140		Intermit/Ephemeral Stream, Wide [River/Stream]	Area	F-5	RRC	9	Railroad/Road Drainage Category
				F-6	HYC	6	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	181-50,000 (T)	Gap Width (Decimeters)
				F-38	GWD	1421-50,000 (P)	Gap Width (Decimeters)
2H140	(T)	Perennial Stream, Narrow [River/Stream]	Line	F-5	RRC	4	Railroad/Road Drainage Category
				F-6	HYC	8	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank

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Surface Drainage (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFT Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2H140	(T)	Perennial Stream, Narrow (Continued)		F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	0-45	Gap Width (Decimeters)
2H140		Perennial Stream, Medium [River/Stream]	Line	F-5	RRC	7	Railroad/Road Drainage Category
				F-6	HYC	8	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	46-180 (T)	Gap Width (Decimeters)
				F-38	GWD	181-1420 (P)	Gap Width (Decimeters)
2H140		Perennial Stream, Wide [River/Stream]	Area	F-5	RRC	9	Railroad/Road Drainage Category
				F-6	HYC	8	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation

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Surface Drainage (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2H140		Perennial Stream, Wide (Continued)		F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	181-50,000 (T)	Gap Width (Decimeters)
				F-38	GWD	1421-50000 (P)	Gap Width (Decimeters)
2H140	(T)	Stream Subject to Tidal Fluctuations, Narrow [River/Stream]	Line	F-5	RRC	4	Railroad/Road Drainage Category
				F-6	HYC	10	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	0-45	Gap Width (Decimeters)
2H140		Stream Subject to Tidal Fluctuations, Medium [River/Stream]	Line	F-5	RRC	7	Railroad/Road Drainage Category
				F-6	HYC	10	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank

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Surface Drainage (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2H140		Stream Subject to Tidal Fluctuations, Medium (Continued)		F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	46-180 (T)	Gap Width (Decimeters)
				F-38	GWD	181-1420	(P) Gap Width (Decimeters)
2H140		Stream Subject to Tidal Fluctuations, Wide [River/Stream]	Area	F-5	RRC	9	Railroad/Road Drainage Category
				F-6	HYC	10	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	181-50,000 (T)	Gap Width (Decimeters)
				F-38	GWD	1421-50000 (P)	Gap Width (Decimeters)
2H140	(T)	Braided Streams, Narrow [River/Stream]	Line	F-5	RRC	4	Railroad/Road Drainage Category
				F-6	HYC	14	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank

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Surface Drainage (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2H140	(T)	Braided Streams, Narrow (Continued)		F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	0-45	Gap Width (Decimeters)
2H140		Braided Streams, Medium [River/Stream]	Line	F-5	RRC	7	Railroad/Road Drainage Category
				F-6	HYC	14	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	46-180 (T)	Gap Width (Decimeters)
				F-38	GWD	181-1420 (P)	Gap Width (Decimeters)
2H140		Braided Streams, Wide [River/Stream]	Area	F-5	RRC	9	Railroad/Road Drainage Category
				F-6	HYC	14	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category

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F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2H140		Braided Streams, Wide (Continued)		F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	181-50,000 (T)	Gap Width (Decimeters)
				F-38	GWD	1421-50,000 (P)	Gap Width (Decimeters)
2H140	(T)	Gorge (Narrow) [Same]	Line	F-5	RRC	4	Railroad/Road Drainage Category
				F-6	HYC	11	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	0-45	Gap Width (Decimeters)
2H140		Gorge (Medium) [Same]	Line	F-5	RRC	7	Railroad/Road Drainage Category
				F-6	HYC	11	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation

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Surface Drainage (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
2H140	(T)	Gorge (Medium) (Continued)		F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	46-180 (T)	Gap Width (Decimeters)
				F-38	GWD	181-1420 (P)	Gap Width (Decimeters)
2H140		Gorge [Wide] [Same]	Area	F-5	RRC	9	Railroad/Road Drainage Category
				F-6	HYC	11	Hydrographic Category
				F-15	OVC	4	Overlay Category
				F-16	WVA	0,1,2	Water Velocity
				F-17	WDA	0-4 (T)	Water Depth Average
				F-17	WDA	0,4,5,6 (P)	Water Depth Average
				F-18	MCC	0,5,14,35,57 66,69,76 (T)	Material Composition Category
				F-18	MCC	0,5,14,57 66,69 (P)	Material Composition Category
				F-19	SBV	0,1,2,3,4	Stream Bank Vegetation
				F-25	BGR	0-998	Bank Gradient-Right Bank
				F-26	BGL	0-998	Bank Gradient-Left Bank
				F-36	BHR	0-9998	Bank Height Cat.-Right Bank
				F-37	BHL	0-9998	Bank Height Cat.-Left Bank
				F-38	GWD	181-50,000 (T)	Gap Width (Decimeters)
				F-38	GWD	1421-50,000 (P)	Gap Width (Decimeters)
2I020		Dam * [Same]	Point	F-2	MCC**	0,18,23,86	Material Composition Category
				F-8	EXS**	0,1,5	Existence Category
				F-10	HGT	0,3,5-998	Height of Feature (Meters)
				F-12	WID**	0-100	Width (Meters)
				F-15	OVC	4	Overlay Category
				F-38	LEN**	0-99(T)	Length/Diameter of Feature
				F-38	LEN**	0-499(P)	Length/Diameter of Feature

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Surface Drainage (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
21020		Dam* [Same] (Continued)	Line	F-2	MCC**	0,18,23,86	Material Composition Category
				F-8	EXS**	0,1,5	Existence Category
				F-10	HGT	0,3,5-998	Height of Feature (Meters)
				F-12	WID**	0-100	Width (Meters)
				F-15	OVC	4	Overlay Category
				F-38	LEN**	0,100-99998(T)	Length/Diameter of Feature
				F-38	LEN**	0,500-99998(P)	Length/Diameter of Feature
		* This feature not collected for PTADB if HGT < 5 meters				** Attribute not collected on TTADB if HGT < 5 meters	
21030		Lock [Same]	Point	F-8	EXS	0,1,5	Existence Category
				F-12	WID	0-100	Width (Meters)
				F-15	OVC	4	Overlay Category
				F-38	LEN	0-99(T)	Length/Diameter of Feature
				F-38	LEN	0-499(P)	Length/Diameter of Feature
			Line	F-8	EXS	0,1,5	Existence Category
				F-12	WID	0-100	Width (Meters)
				F-15	OVC	4	Overlay Category
				F-38	LEN	0,100-99998(T)	Length/Diameter of Feature
				F-38	LEN	0,500-99998(P)	Length/Diameter of Feature
9D010*		Miscellaneous Surface Drainage Feature [Miscellaneous Graphic Features]	Point	F-15	OVC	4	Overlay Category
			Line	F-15	OVC	4	Overlay Category
			Area	F-15	OVC	4	Overlay Category

* In the ITD/PITD SLF text record enter the complete feature description and dimensional value for all characteristics of the Miscellaneous surface drainage features.

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F Code ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1N010	Single Track, Narrow Gauge [Railroad Tracks]	Line	F-4	RRA	1,5	Railroad Attributes
			F-5	RRC	4	Road/Railroad Categories
			F-6	LTC	4	Lane/Track Characteristics
			F-8	EXS	1,5	Existence Category
			F-15	OVC	5	Overlay Category
1N010	Single Track, Normal Gauge [Railroad Tracks]	Line	F-4	RRA	1,5	Railroad Attributes
			F-5	RRC	5	Road/Railroad Categories
			F-6	LTC	4	Lane/Track Characteristics
			F-8	EXS	1,5	Existence Category
			F-15	OVC	5	Overlay Category
1N010	Single Track, Broad Gauge [Railroad Tracks]	Line	F-4	RRA	1,5	Railroad Attributes
			F-5	RRC	1	Road/Railroad Categories
			F-6	LTC	4	Lane/Track Characteristics
			F-8	EXS	1,5	Existence Category
			F-15	OVC	5	Overlay Category
1N010	Multiple Track, Narrow Gauge [Railroad Tracks]	Line	F-4	RRA	1,5	Railroad Attributes
			F-5	RRC	4	Road/Railroad Categories
			F-6	LTC	3	Lane/Track Characteristics
			F-8	EXS	1,5	Existence Category
			F-15	OVC	5	Overlay Category
1N010	Multiple Track, Normal Gauge [Railroad Tracks]	Line	F-4	RRA	1,5	Railroad Attributes
			F-5	RRC	5	Road/Railroad Categories
			F-6	LTC	3	Lane/Track Characteristics
			F-8	EXS	1,5	Existence Category
			F-15	OVC	5	Overlay Category

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1N010		Multiple Track, Broad Gauge [Railroad Tracks]	Line	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	1	Road/Railroad Categories
				F-6	LTC	3	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
1N010		Dismantled Railroad [Railroad Tracks]	Line	F-8	EXS	8	Existence Category
				F-15	OVC	5	Overlay Category
1N030	(P)	Passing Track, Narrow Gauge [Railroad Passing]	Point	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	4	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature
	(T)		Line	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	4	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature

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F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1N030	(P)	Passing Track, Normal Gauge [Railroad Passing]	Point	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	5	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature
	(T)		Line	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	5	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature
1N030	(P)	Passing Track, Broad Gauge [Railroad Passing]	Point	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	1	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature
	(T)		Line	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	1	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,0000	Length/Diameter of Feature

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1N050	(P)	Siding Track, Narrow Gauge [Railroad Siding]	Point	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	4	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature
	(T)		Line	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	4	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature
1N050	(P)	Siding Track, Normal Gauge [Railroad Siding]	Point	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	5	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature
	(T)		Line	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	5	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1N050	(P)	Siding Track, Broad Gauge [Railroad Siding]	Point	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	1	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature
	(T)		Line	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	1	Road/Railroad Categories
				F-6	LTC	4	Lane/Track Characteristics
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	280-20,000	Length/Diameter of Feature
1N080	(P)	Rail Yard, Narrow Gauge [Railroad Yard]	Point	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	4	Road/Railroad Categories
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	0-99998	Length/Diameter of Feature
			Line	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	4	Road/Railroad Categories
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	0-99998	Length/Diameter of Feature
	(T)		Area	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	4	Road/Railroad Categories
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	0-99998	Length/Diameter of Feature

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1N080	(P)	Rail Yard, Normal Gauge [Railroad Yard]	Point	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	5	Road/Railroad Categories
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	0-99998	Length/Diameter of Feature
			Line	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	5	Road/Railroad Categories
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	0-99998	Length/Diameter of Feature
	(T)		Area	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	5	Road/Railroad Categories
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	0-99998	Length/Diameter of Feature

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1N080	(P)	Rail Yard, Broad Gauge [Railroad Yard]	Point	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	1	Road/Railroad Categories
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	0-99998	Length/Diameter of Feature
			Line	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	1	Road/Railroad Categories
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	0-99998	Length/Diameter of Feature
	(T)		Area	F-4	RRA	1,5	Railroad Attributes
				F-5	RRC	1	Road/Railroad Categories
				F-8	EXS	1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-38	LEN	0-99998	Length/Diameter of Feature
1P010		Cart Track [Same]	Line	F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
1P030		All Weather Hard Surface Highway [Road]	Line	F-2	RSC	0,1,6	Road/RR Structure Category
				F-4	RST	1	Road/Runway Surface Type
				F-5	WTC	1	Weather Type Category
				F-7	TWC	1,2,3	Travelway Characteristics
				F-8	EXS	0,1,5	Existence Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
				F-16	SGC	0-98	Slope/Gradient Category
				F-24	WDD	0-500	Width-Decimeters

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFT Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1P030		All Weather Loose Surface Highway [Road]	Line	F-2	RSC	0,1,6	Road/RR Structure Category
				F-4	RST	2	Road/Runway Surface Type
				F-5	WTC	1	Weather Type Category
				F-7	TWC	3	Travelway Characteristics
				F-8	EXS	0,1,5	Existence Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
				F-16	SGC	0-98	Slope/Gradient Category
				F-24	WDD	0-500	Width-Decimeters
1P030		Fair Weather Loose Surface Highway [Road]	Line	F-2	RSC	0,1,6	Road/RR Structure Category
				F-4	RST	2	Road/Runway Surface Type
				F-5	WTC	2	Weather Type Category
				F-7	TWC	3	Travelway Characteristics
				F-8	EXS	0,1,5	Existence Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
				F-16	SGC	0-98	Slope/Gradient Category
				F-24	WDD	0-500	Width-Decimeters
1Q040		Road Bridge [Bridge]	Point	F-3	TUC	4	Transportation Use Category
				F-4	BCC	0-3	Bypass Condition Category
				F-5	NOS	0-98	Number of Spans
				F-8	EXS	0,1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-23	UBD	0-998	Underbridge Clearance-Decimeters
				F-24	WDD	0-500	Width-Decimeters
				F-25	OHD	0-501	Overhead Clearance-Decimeters
				F-28	LC1	0-200	Load Class Type:One-way Wheeled
				F-29	LC2	0-200	Load Class Type:Two-way Wheeled
				F-30	LC3	0-200	Load Class Type:One-way Tracked
				F-31	LC4	0-200	Load Class Type:Two-way Tracked
		(Continued)					

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1Q040		Road Bridge [Bridge] (Continued)		F-36	BRN	1-9998	Bridge Reference Number
				F-38	LND	0-999 (T)	Length-Decimeters
				F-38	LND	0-4999 (P)	Length-Decimeters
			Line	F-3	TUC	4	Transportation Use Category
				F-4	BCC	0-3	Bypass Condition Category
				F-5	NOS	0-98	Number of Spans
				F-8	EXS	0,1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-23	UBD	0-998	Underbridge Clearance-Decimeters
				F-24	WDD	0-500	Width-Decimeters
				F-25	OHD	0-501	Overhead Clearance-Decimeters
				F-28	LC1	0-200	Load Class Type:One-way Wheeled
				F-29	LC2	0-200	Load Class Type:Two-way Wheeled
				F-30	LC3	0-200	Load Class Type:One-way Tracked
				F-31	LC4	0-200	Load Class Type:Two-way Tracked
				F-36	BRN	1-9998	Bridge Reference Number
				F-38	LND	1000-99998 (T)	Length-Decimeters
				F-38	LND	5000-99998 (P)	Length-Decimeters
1Q040		Railroad Bridge [Bridge]	Point	F-3	TUC	3	Transportation Use Category
				F-8	EXS	0,1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-25	OHD	0-501	Overhead Clearance-Decimeters
				F-38	LND	0-999 (T)	Length-Decimeters
				F-38	LND	0-4999 (P)	Length-Decimeters
			Line	F-3	TUC	3	Transportation Use Category
				F-8	EXS	0,1,5	Existence Category
				F-15	OVC	5	Overlay Category
				F-25	OHD	0-501	Overhead Clearance-Decimeters
				F-38	LND	1000-99998 (T)	Length-Decimeters
				F-38	LND	5000-99998 (P)	Length-Decimeters

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1Q045		Bridge Span [Same]	Point	F-3	MCC	0,18,48,60, 65,83,86,97	Material Composition Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
				F-36	BRN	1-9998	Bridge Reference Number
				F-38	LND	0,1-999 (T)	Length-Decimeters
				F-38	LND	0,1-4999 (P)	Length-Decimeters
			Line	F-3	MCC	0,18,48,60, 65,83,86,97	Material Composition Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
				F-36	BRN	1-9998	Bridge Reference Number
				F-38	LND	0,1000-99998(T)	Length-Decimeters
				F-38	LND	0,5000-99998(P)	Length-Decimeters
1Q058		Constriction [Same]	Point	F-15	OVC	5	Overlay Category
				F-24	WDD	0-40	Width-Decimeters
1Q068		Drop Gate Road [Drop Gate]	Point	F-3	TUC	4	Transportation Use Category
				F-15	OVC	5	Overlay Category
1Q068		Drop Gate Railroad [Drop Gate]	Point	F-3	TUC	3	Transportation Use Category
				F-15	OVC	5	Overlay Category
1Q070		Ferry, Road [Ferry Crossing]	Point	F-3	TUC	4	Transportation Use Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
			Line	F-3	TUC	4	Transportation Use Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1Q070		Ferry, Railroad [Ferry Crossing]	Point	F-3	TUC	3	Transportation Use Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
			Line	F-3	TUC	3	Transportation Use Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
1Q118		Road Radius of Curvature [Same]	Point	F-15	OVC	5	Overlay Category
1Q130		Tunnel, Road [Tunnel/Tunnel Entrance/Exit]	Point	F-3	TUC	4	Transportation Use Category
				F-8	EXS	0,1,5	Existence Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
				F-24	WDD	0-500	Width-Decimeters
				F-25	OHD	0-500	Overhead Clearance-Decimeters
				F-38	LEN	0-99 (T)	Length/Diameter of Feature
				F-38	LEN	0-499 (P)	Length/Diameter of Feature
			Line	F-3	TUC	4	Transportation Use Category
				F-8	EXS	0,1,5	Existence Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
				F-24	WDD	0-500	Width-Decimeters
				F-25	OHD	0-500	Overhead Clearance-Decimeters
				F-38	LEN	100-42,000 (T)	Length/Diameter of Feature
				F-38	LEN	500-42,000 (P)	Length/Diameter of Feature

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1Q130		Tunnel, Railroad [Tunnel/Tunnel Entrance/Exit]	Point	F-3	TUC	3	Transportation Use Category
				F-8	EXS	0,1,5	Existence Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
				F-24	WDD	0-500	Width-Decimeters
				F-25	OHD	0-500	Overhead Clearance-Decimeters
				F-38	LEN	0-99 (T)	Length/Diameter of Feature
				F-38	LEN	0-499 (P)	Length/Diameter of Feature
			Line	F-3	TUC	3	Transportation Use Category
				F-8	EXS	0,1,5	Existence Category
				F-9	ACC	0,1,2	Accuracy Category
				F-15	OVC	5	Overlay Category
				F-24	WDD	0-500	Width-Decimeters
				F-25	OHD	0-500	Overhead Clearance-Decimeters
				F-38	LEN	100-20,000 (T)	Length/Diameter of Feature
				F-38	LEN	500-20,000 (P)	Length/Diameter of Feature
1U160		Airfield-Hard/Paved [Runway]	Line	F-2	DLA	2	Definition of Landing Area
				F-5	RST	1	Road/Runway Surface Type
				F-8	EXS	0,1,5,6	Existence Category
				F-12	WID	0-300	Width (Meters)
				F-15	OVC	5	Overlay Category
				F-35	LEN	0-5000	Length/Diameter of Feature
	(T)		Area	F-2	DLA	2	Definition of Landing Area
				F-5	RST	1	Road/Runway Surface Type
				F-8	EXS	0,1,5,6	Existence Category
				F-12	WID	0-300	Width (Meters)
				F-15	OVC	5	Overlay Category
				F-35	LEN	0-5000	Length/Diameter of Feature

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1U160		Airfield-Loose/Unpaved [Runway]	Line	F-2	DLA	0,1,2	Definition of Landing Area
				F-5	RST	2	Road/Runway Surface Type
				F-8	EXS	0,1,5,6	Existence Category
				F-12	WID	0-300	Width (Meters)
				F-15	OVC	5	Overlay Category
				F-35	LEN	0-5000	Length/Diameter of Feature
	(T)		Area	F-2	DLA	0,1,2	Definition of Landing Area
				F-5	RST	2	Road/Runway Surface Type
				F-8	EXS	0,1,5,6	Existence Category
				F-12	WID	0-300	Width (Meters)
				F-15	OVC	5	Overlay Category
				F-35	LEN	0-5000	Length/Diameter of Feature
1U160	(P)	Landing Area Hard/Paved [Runway]	Point	F-2	DLA	1	Definition of Landing Area
				F-5	RST	1	Road/Runway Surface Type
				F-15	OVC	5	Overlay Category
				F-35	LEN	0-5000	Length/Diameter of Feature
				F-36	WID	0-5000	Width (Meters)
1U160	(P)	Landing Area Loose/Unpaved [Runway]	Point	F-2	DLA	1	Definition of Landing Area
				F-5	RST	2	Road/Runway Surface Type
				F-15	OVC	5	Overlay Category
				F-35	LEN	0-5000	Length/Diameter of Feature
				F-36	WID	0-5000	Width (Meters)
2H070		On Route Ford [Ford]	Point	F-15	OVC	5	Overlay Category
			Line	F-15	OVC	5	Overlay Category

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Transportation (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
9D010*		Miscellaneous Transportation Feature [Miscellaneous Graphic Features]	Point	F-15	OVC	5	Overlay Category
			Line	F-15	OVC	5	Overlay Category
			Area	F-15	OVC	5	Overlay Category

* In the ITD/PITD SLF text record enter the complete feature description and dimensional values for all characteristics of the Miscellaneous Transportation features.

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Section 6 OBSTACLES

F Code	ITD(T) PITD(P)	Feature Name [DMAFF Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
1L060		Dragon Teeth [Same]	Line	F-15	OVC	6	Overlay Category
			Area	F-15	OVC	6	Overlay Category
1L160		Pipeline [Same]	Line	F-3	LOC	0,3,4	Location/Origin Category
				F-15	OVC	6	Overlay Category
1L260	(T)	Wall/Fence [Wall]	Line	F-15	OVC	6	Overlay Category
2B070		Volcanic Dike [Dike]	Line	F-9	MCC	94	Material Composition Category
				F-15	OVC	6	Overlay Category
2B220		Crossing Point [Ramp]	Point	F-7	HLC	19	Hydrographic Location Category
				F-15	OVC	6	Overlay Category
2H100	(T)	Moat [Same]	Line	F-15	OVC	6	Overlay Category
4B010		Escarpment [Bluff/Cliff/ Escarpment]	Line	F-15	OVC	6	Overlay Category
4B070		Road/RR Cut [Cut]	Line	F-15	OVC	6	Overlay Category
4B080		Depression [Same]	Area	F-15	OVC	6	Overlay Category

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Obstacles (Continued)

F Code	ITD(T) PITD(P)	Feature Name [DMAFT Feature Name]	F Type	F At. No.	At. Code	Values	Attribute
4B090		Embankment [Same]	Line	F-15	OVC	6	Overlay Category
4B120		Road/RR Fill [Fill]	Line	F-15	OVC	6	Overlay Category
5A020	(T)	Hedgerow [Same]	Line	F-15	OVC	6	Overlay Category
9D010*		Miscellaneous Obstacle	Point	F-15	OVC	6	Overlay Category
		Feature [Miscellaneous Graphic Features]	Line	F-15	OVC	6	Overlay Category
			Area	F-15	OVC	6	Overlay Category

* In the ITD/PITD SLF text record enter the complete feature description and dimensional values for all characteristics of the Miscellaneous Obstacle features.

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APPENDIX B

ITD ATTRIBUTE LISTING

B.1 SCOPE

B.1.1 Scope. This appendix provides a guide to the ITD attribute codes, attribute values, and value meanings. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

B.3 ITD ATTRIBUTE CODE NAMES AND ATTRIBUTE VALUES

B.3.1 ITD attribute table.

Attribute Code	Attribute Values	Value Meaning
ACC		Accuracy Category
	0	Unknown
	1	Accurate
	2	Approximate
BCC		Bypass Condition Category
	0	Unknown
	1	Difficult
	2	Easy
	3	Impossible
BDC		Brushland Density Category
	0	Unknown
	1	Open to Medium (0-50% Coverage)
	2	Medium to Dense (51-100% Coverage)
BGL		Bank Gradient (Slope) Category-Left Bank
	0	Unknown
	1	1%
	.	
	.	
	.	
	998	998%
BGR		Bank Gradient (Slope) Category-Right Bank
		Same Values As BGL
BHL		Bank Height Category-Left Bank
	0	Unknown
	1	1 Decimeter
	.	
	.	
	.	
	9998	9998 Decimeter

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APPENDIX B

ITD attribute code name and attribute values (Continued)

Attribute Code	Attribute Values	Value Meaning
BHR	Bank Height Category-Right Bank	
	Same values as BHL	
BRN	Bridge Reference Number	
	1	Bridge number 1
	.	
	.	
	.	
	9998	Bridge number 9998
DLA	Definition of Landing Area	
	0	Unknown
	1	No well defined runway
	2	Well defined runway
DMT	Density Measure (% of Tree/Canopy Cover)	
	0	Unknown
	1	1%
	.	
	.	
	.	
	100	100%
EXS	Existence Category	
	0	Unknown
	1	Definite
	5	Under Construction
	6	Abandoned/non-operational
	8	Dismantled
GSC	Ground Slope Category	
	0	Unknown
	1	0-3%
	2	>3-10%
	3	>10-20%
	4	>20-30%
	5	>30-45%
	6	>45%
	7	0->45% (Naturally and/or culturally dissected land).
GWD	Gap Width Decimeters	
	0	Unknown
	1	1 Decimeter
	.	
	.	
	.	
	99998	99998 Decimeters

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APPENDIX B

ITD attribute code name and attribute values (Continued)

Attribute Code	Attribute Values	Value Meaning
HGT		Height of Feature Above Ground Level
	0	Unknown
	1	1 Meter
	.	
	.	
	.	
	998	998 Meters
HLC		Hydrographic Location Category
	0	Unknown
	19	Above Surface
HYC		Hydrographic Category
	6	Non-Perennial/Intermittent/Fluctuating and Ephemeral
	8	Perennial/Permanent
	10	Tidal/Tidal Fluctuation
	11	Steep Sides
	14	Braided
LC1		Load Class Type: One-Way, Wheeled Vehicles
	0	Unknown
	1	1 Short Ton
	.	
	.	
	.	
	200	200 Short Tons
LC2		Load Class Type: Two-way, Wheeled Vehicles
	0	Unknown
	1	1 Short Ton
	.	
	.	
	.	
	200	200 Short Tons
LC3		Load Class Type:One-way, Tracked Vehicles
	0	Unknown
	1	1 Short Ton
	.	
	.	
	.	
	200	200 Short Tons

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APPENDIX B

ITD attribute code name and attribute values (Continued)

Attribute Code	Attribute Values	Value Meaning
LC4		Load Class Type:Two-way, Tracked Vehicles
	0	Unknown
	1	1 Short Ton
	.	
	.	
	.	
	200	200 Short Tons
LEN		Length/Diameter of Feature
	0	Unknown
	1	1 Meter
	.	
	.	
	.	
	99998	99998 Meters
LND		Length in Decimeter
	0	Unknown
	1	1 Decimeter
	.	
	.	
	.	
	99998	99998 Decimeters
LOC		Location/Origin Category
	0	Unknown
	1	Below Ground Level
	3	On Ground Surface
	4	Suspended or Elevated
LTC		Lane/Track Characteristics
	3	Multiple
	4	Single
MCC		Material Composition Category
	0	Unknown
	4	Bare/Cleared
	5	Bedrock
	14	Clay
	18	Concrete
	23	Earthwork
	24	Evaporites
	35	Gravel
	48	Masonry (Stone/Brick)
	57	Paved
	60	Prestressed Concrete

(Continued)

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APPENDIX B

ITD attribute code name and attribute values (Continued)

Attribute Code	Attribute Values	Value Meaning
MCC		Material Composition Category (Continued)
	65	Reinforced Concrete
	66	Rock, Rocky
	69	Sand
	76	Silt
	77	Soil
	83	Steel
	86	Stone
	94	Volcanic
	97	Wood
NOS		Number of Spans
	0	Unknown
	1-98	(max. 2 digits)
OHD		Overhead Clearance-Decimeters
	0	Unknown
	1	1 Decimeter
	.	
	.	
	.	
	500	500 Decimeters
	501	Unlimited
OVC		Overlay Category
	0	Unknown
	1	Surface Configuration
	2	Vegetation
	3	Surface Materials
	4	Surface Drainage
	5	Transportation
	6	Obstacles
RRA		Railroad Attributes
	1	Electrified
	5	Non-electrified
RRC		Railroad/Road Categories (For ITD, RRC is used for some Surface Drainage, as well as some Transportation Features)
	1	Broad Gauge
	4	Narrow/Narrow Gauge
	5	Normal (Standard) Gauge
	7	Medium
	9	Wide

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APPENDIX B

ITD attribute code name and attribute values (Continued)

Attribute Code	Attribute Values	Value Meaning
RSC		Road/RR Structure Category
	0	Unknown
	1	Non-elevated
	6	Elevated on Structure
RST		Road/Runway Surface Type
	1	Hard/Paved
	2	Loose/Unpaved
SBV		Stream Bank Vegetation
	0	Unknown
	1	Dense Vegetation on the right bank
	2	Dense Vegetation on the left bank
	3	Dense Vegetation on both banks
	4	Neither bank contains dense vegetation
SDC		Soil Depth Category
	0	Unknown
	1	>= 0.5 meters
	2	< 0.5 meters
SDS		Stem Diameter Size
	0	Unknown
	1	1 cm
	.	
	.	
	.	
	900	900 cm
SGC		Slope/Gradient Category
	0	Unknown
	1	0 - <2%
	2	2
	3	3
	.	
	.	
	.	
	98	98%
SRQ		Surface Roughness Qualifier
	0	No Data (Unknown) (Predefined for PTADB and TTADB)
	1	No Surface Roughness effect (Predefined for PTADB and TTADB)
	2	Area of high landslide potential (Predefined for TTADB)
	3-98	Unique descriptions tailored to individual project areas

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APPENDIX B

ITD attribute code name and attribute values (Continued)

Attribute Code	Attribute Values	Value Meaning
STC	Soil Type Category	
	0	Unknown
	1	GW - Well-graded gravels, gravel-sand mixtures, little or no fines.
	2	GP - Poorly graded gravels or gravel-sand mixtures, little or no fines.
	3	GM - Silty gravels, gravel-sand-silt mixtures.
	4	GC - Clayey gravels, gravel-sand-clay mixtures.
	5	SW - Well-graded sand, gravelly sands, little or no fines.
	6	SP - Poorly graded sands or gravelly sands, little or no fines.
	7	SM - Silty sands, sand-silt mixtures.
	8	SC - Clayey sands, sand-clay mixtures.
	9	ML - Inorganic silts and very fine sands, rock flour, silty or clayey fine sands or clayey silts with slight plasticity.
	10	CL - Inorganic clays of low to medium plasticity, gravelly clays, sandy clays, silty clays, lean clays.
	11	OL - Organic silts and organic silty clays of low plasticity.
	12	CH - Inorganic clays of high plasticity, fat clays.
	13	MH - Inorganic silts, micaceous or diatomaceous fine sandy or silty soils, elastic silts.
	14	OH - Organic clays of medium to high plasticity, organic silts.
	15	PT - Peat and other highly organic soils.
SWC	Soil Wetness Category	
	0	Unknown
	1	Dry
	2	Moist
	3	Wet
TSD	Tree Spacing Category	
	0	Unknown
	1	1 Decimeter
	.	
	.	
	.	
	500	500 Decimeters
TUC	Transportation Use Category	
	3	Railroad
	4	Road

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APPENDIX B

ITD attribute code name and attribute values (Continued)

Attribute Code	Attribute Values	Value Meaning
TWC		Travelway Characteristics
	1	Travelway for Dual/Divided Same Widths
	2	Travelway for Dual/Divided Different Widths
	3	Non-divided
UBD		Underbridge Clearance-Decimeters
	0	Unknown
	1	1 Decimeter
	.	
	.	
	.	
	998	998 Decimeters
UGD		Undergrowth Density Category
	0	Unknown
	1	None to sparse
	2	Medium to Dense
VEG		Vegetation Characteristics
	0	Unknown
	1	Dry Crops
	2	Shifting (cultivation/usage)
	3	Terraced
	4	Rice Paddy
	5	Agriculture With Scattered Forests
	8	Grassland
	9	Grassland w/Scatt. Trees & Scrub Growth
	13	Deciduous
	14	Evergreen
	15	Mixed
	17	Palm
	19	Mangrove
	24	Forest Clearing
WDA		Water Depth Average
	0	Unknown
	1	<=0.8 meters
	2	>0.8 - 1.6 m
	3	>1.6 - 2.4 m
	4	>2.4 m
	5	<=1.2 m
	6	>1.2 m - 2.4 m

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APPENDIX B

ITD attribute code name and attribute values (Continued)

Attribute Code	Attribute Values	Value Meaning
WDD	Width-Decimeters	
	0	Unknown
	1	1 Decimeter
	.	
	.	
	.	
	500	500 Decimeters
WID	Width	
	0	Unknown
	1	1 Meter
	.	
	.	
	.	
	998	998 Meters
WTC	Weather Type Category	
	1	All weather
	2	Fair/Dry Weather
WVA	Water Velocity Average	
	0	Unknown
	1	<=1.5 m/sec.
	2	>1.5 m/sec.

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CONCLUDING MATERIAL

Custodians:
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Army -
Navy - NO

Review activities:
Marine Corps - MC

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DMA - MP

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-89014A	2. DOCUMENT DATE (YYMMDD) 960315

3. DOCUMENT TITLE PERFORMANCE SPECIFICATION
INTERIM TERRAIN DATA (ITD)/PLANNING INTERIM TERRAIN DATA (PITD)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

5. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME DEFENSE MAPPING AGENCY	b. TELEPHONE (Include Area Code) (1) Commercial (703) 275-8509	(2) AUTOVON 235-8509
c. ADDRESS (Include Zip Code) ATTN: ATIS 8613 Lee Highway, Fairfax, VA 22031-2137	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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